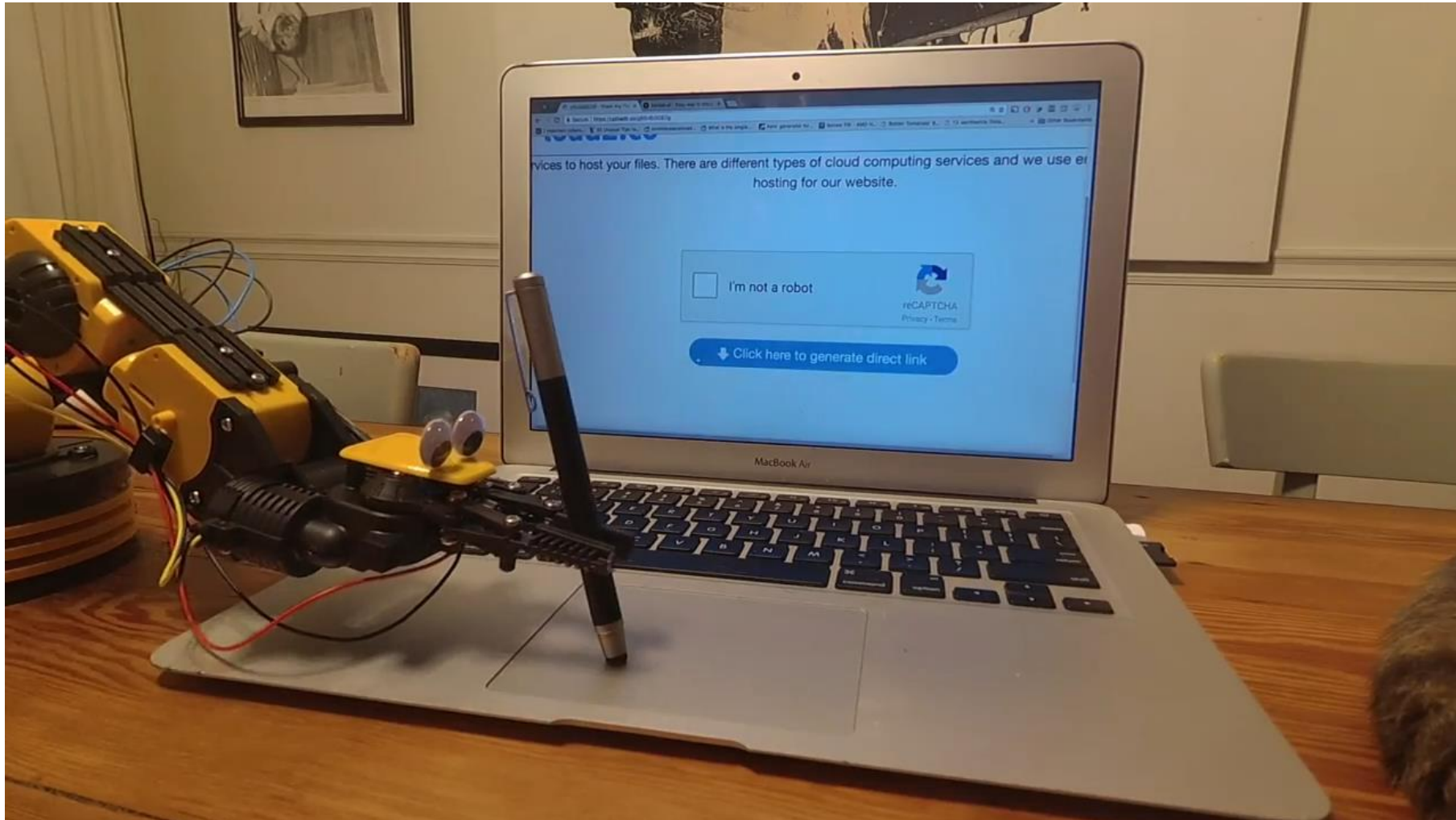


Benchmarking Cognition?



<https://www.youtube.com/watch?v=fsF7enQY8ul>

Turing Test

The Turing test was proposed by Alan M. Turing in 1950 to determine whether a computer can think. It is based on an **imitation game**:

- An **interrogator** can communicate with a player A (**a man**) and a player B (**a woman**) through **written notes**
- The interrogator tries to determine whether player A or player B is the woman
- In the test, player A is **replaced by a computer** that tries to make the interrogator believe that it is the woman
- The computer passes the test when it manages to deceive the interrogator **as often as the man**

In the more common standard interpretation of the test, the computer only **imitates a human** and the interrogator is not deceived but only needs to tell computer and human apart.

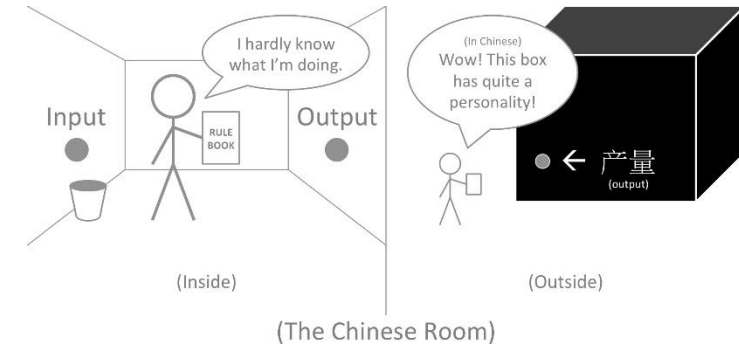
From https://en.wikipedia.org/wiki/Turing_test

Chinese Room Argument

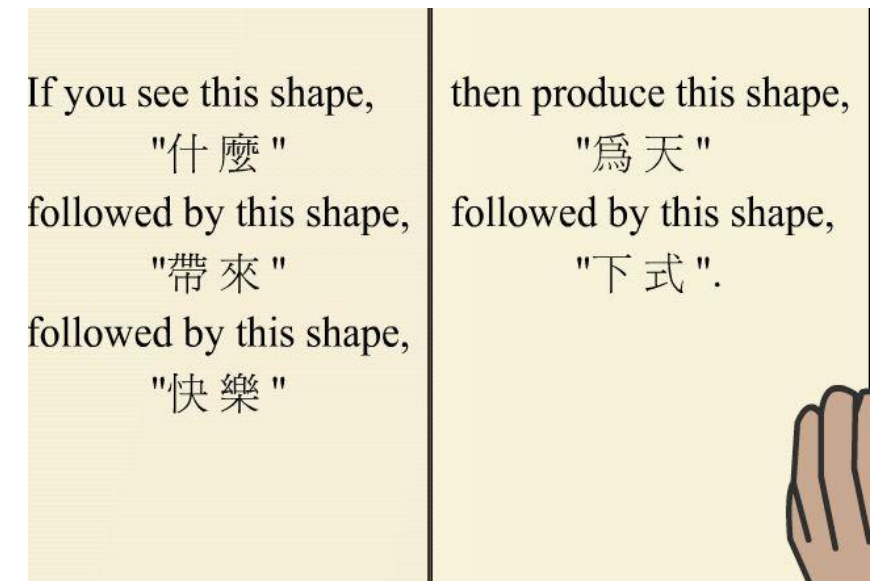
The Chinese Room Argument is a thought experiment by philosopher John Searle which states that a computer cannot think even if it passes the Turing test:

- A person **without any knowledge** of the Chinese language is sitting in a room
- The room contains an **archive with rules** that map Chinese characters to appropriate Chinese responses
- The room contains **a slot** through which the person receives questions from speakers of the Chinese language
- Like a **computer program**, the person uses rule the archive to generate answers to the questions by performing operations on strings
- Even if the person produces perfect answers, he/she still does not understand the **meanings of the characters**

Encyclopædia Britannica



<http://princetonlegree.com/the-living-dead/>

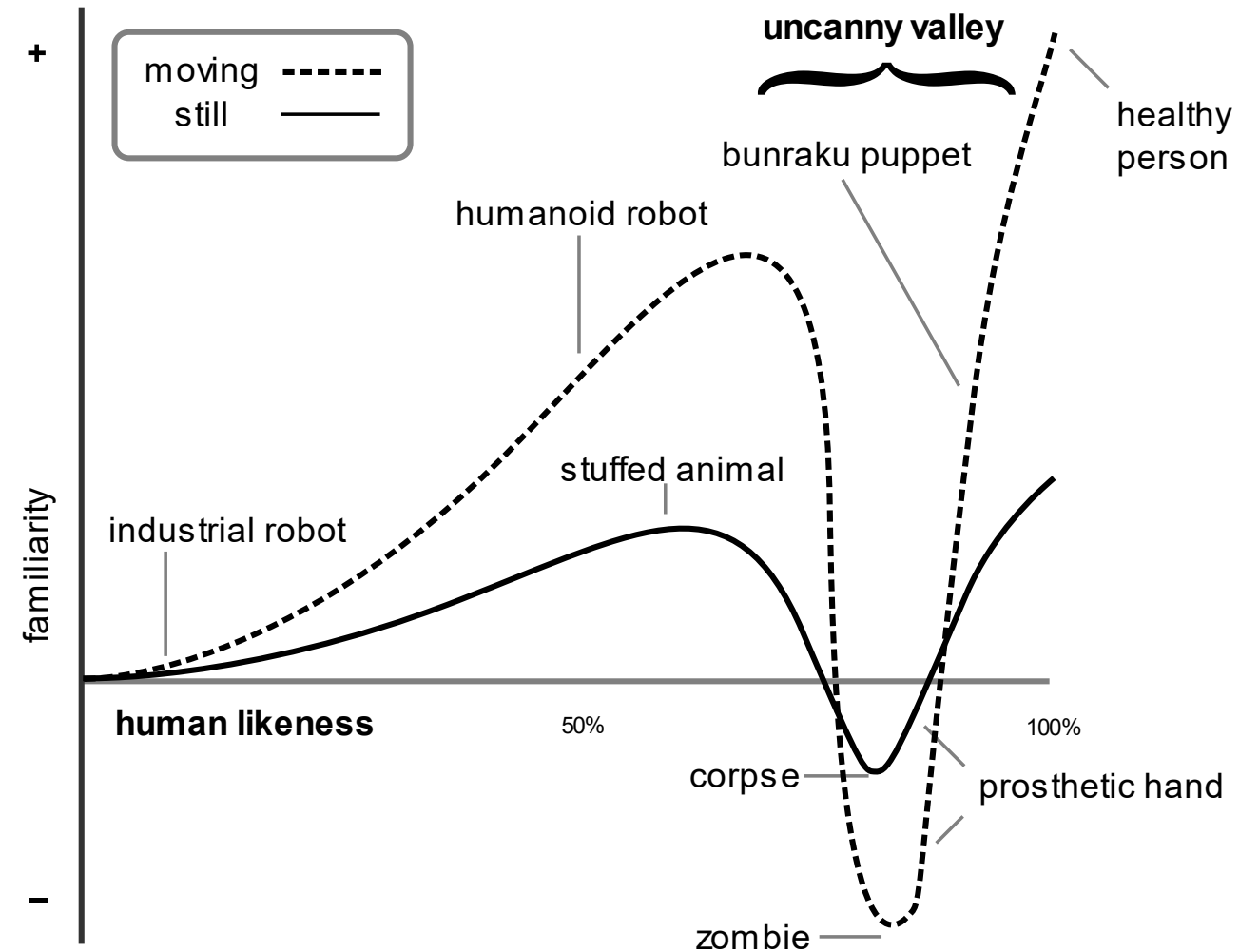


http://www.mind.ilstu.edu/curriculum/searle_chinese_room/searle_chinese_room.php

The Uncanny Valley

The physical appearance of a cognitive system has a huge impact on the interaction with humans.

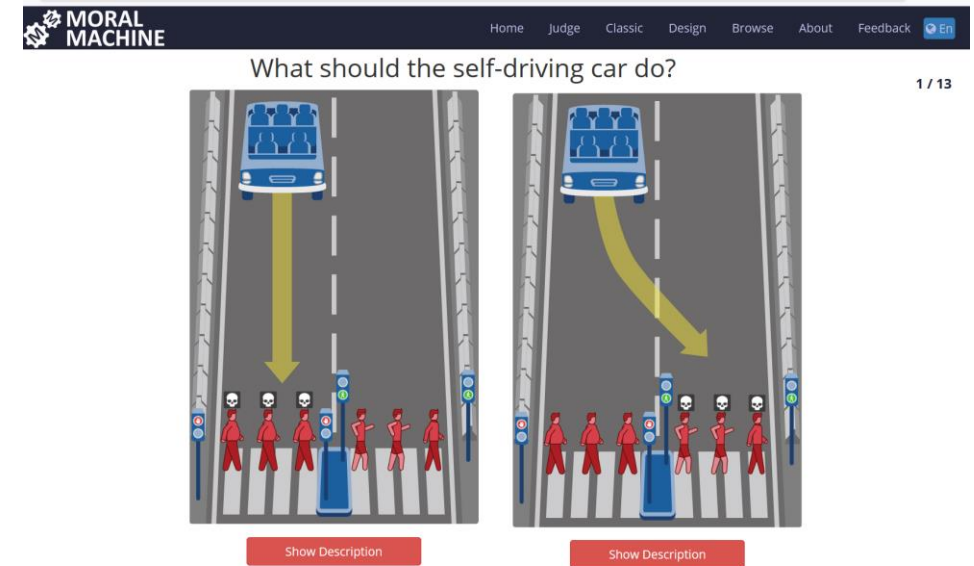
The Uncanny Valley phenomenon states that a steady increase in “human-likeness” of a robot does not yield a steady increase of familiarity to humans



Plot of the Uncanny Valley by Smurrayinchester after Masahiro Mori and Karl MacDorman licensed under CC BY-SA 3.0

Ethical and Legal Aspects

- What about the **liability** of a cognitive system, e.g. an autonomous car?
- How can an artificial cognitive system decide **which person's life to save** in an unavoidable accident?
- Can an artificial cognitive system be **conscious**?
- Does an artificial cognitive system that acts like a human possess **personhood**?
- ...



<http://moralmachine.mit.edu/>

Summary

- Robotics, computer science and engineering in general rely on **formal descriptions** of problems and tasks; the quality of these descriptions determines the performance of the system
- With traditional methods, real world-tasks can only be implemented by making **a priori assumptions** about the world; if these assumptions do not hold, the system fails
- Cognitive systems can act **self-reliantly** and **independently** and therefore do not need complete formal specifications to **achieve a goal**
- Cognition is a **system-wide property** that emerges from the synergy of a repertoire of **cognitive functions**
- A cognitive system is not an isolated entity but is **situated** in a specific **environment** within which it **acts** and **interacts**